

- 1 (a) The electric field lines, along which the electric forces exerted by the charged sphere on other charges act, are always perpendicular to the surface of the sphere (i.e. the electric field lines are radial). M1

Hence the field lines appear to originate from the centre of the sphere. A0

- (b) (i) arrow pointing to the left labelled 'electric force' (F_E) **and** arrow pointing downwards labelled 'weight' (W) C1

(ii) 1.

Electric force on the particle

$$F_E = QE$$

$$F_E = VQ / d$$

$$= (2.0 \times 10^2 \times 8.0 \times 10^{-19}) / 4.0 \times 10^{-2} = 4.0 \times 10^{-15} \text{ N} \quad \text{A1}$$

$$\text{resultant force} = \sqrt{(3.9 \times 10^{-15})^2 + (4.0 \times 10^{-15})^2} = 5.6 \times 10^{-15} \text{ N} \quad \begin{array}{l} \text{C1} \\ \text{A1} \end{array}$$

2. let θ be the angle to horizontal

$\tan \theta = \text{weight} / \text{electric force}$

$$\theta = \tan^{-1} (3.9 \times 10^{-15} / 4.0 \times 10^{-15})$$

$$= 44^\circ$$

Substitution must be shown

A1

- (c) (i) downward sloping line from (0, 2.0)
magnitude of gradient of line increases with time and line ends at (T, 0) M1
A1

(ii) Horizontally:

$$s = \frac{1}{2} at^2$$

$$2.0 \times 10^{-2} = \frac{1}{2} (4.0 \times 10^{-15} / (3.9 \times 10^{-15} / 9.81)) T^2 \quad \text{C1}$$

$$T = 0.063 \text{ s} \quad \text{A1}$$

- 2 (a) (i) gravitational field strength is equal to the negative gravitational potential B1